# Anushka Murthy

anchmurthy.github.io anushkam@stanford.edu LinkedIn

## Education

**Stanford University** *2022 – Present*
*PhD in Mathematics (2022-2023), Operations Research (2023-present)* *Stanford, CA*

- Advisors: Ramesh Johari and Irene Lo.
- Awards: Stanford HAI Data Science Scholarship, NSF Graduate Fellowship, EDGE Fellowship.
- Selected coursework: Graduate Analysis and Algebra, Stochastic Systems, Optimization, Causal Inference, Topics in Social Data, Game Theory, Topics in Market Design.
- Activities: Dean's Graduate Student Advisory Council, Stanford Women in Math Mentoring, mentor in Undergraduate Diversity in Research program, violinist in music department.

**Columbia University School of Engineering and Applied Science** *2018 – 2022*
*Bachelor of Science in Applied Math* *New York, NY*

- Graduated *summa cum laude*, GPA: 4.07/4.33.

## Papers and Interests

**Interests:** My methodological interests are in applied probability, market design, and causal inference. I am particularly interested in the impact of data-driven interventions (A/B experiments, recommender systems, AI agents) on search and learning in marketplaces.

1. *Can Labor Markets Function in the Age of AI? The Evaluation Bottleneck in Hiring*, I. Ashlagi, R. Johari, J. Kleinberg, A. Murthy. Working paper.
2. *One-sided Randomization in Two-sided Platform Experiments: Which Side Should be Randomized?*, R. Johari and A. Murthy. Working paper.
3. *When Does Interference Matter? Decision-Making in Platform Experiments*, R. Johari, H. Li, A. Murthy, G. Weintraub. Major revision at **Management Science**.
4. *Dynamic Contracting for Payment for Ecosystem Services*, I. Lo and A. Murthy. Major revision at **Management Science**.
5. *One-point asymptotics for half-flat ASEP*, E. Dimitrov and A. Murthy. **Annals of Applied Probability**, 34(1B): 1136-1176 (2024).

## Industry Experience

**Amazon Middle Mile Marketplace Science** *June – September 2025*
*Applied Scientist Intern* *Bellevue, WA*

- Developed machine learning models to predict trailer pool adjustment cost in the spot market.
- Validated and enhanced estimation methods for quantifying the causal impact of increasing lane volume through Amazon Freight contracts on trailer pool adjustment costs, enabling more accurate cost forecasting.

## Teaching Experience

**Simons Laufer Mathematical Sciences Institute (MSRI)** *June 2023*
*Teaching Assistant* *Berkeley, CA*

- Assisted in graduate summer school on "Mathematics and Computer Science of Market Design."

**Stanford University** *2022 – Present*
*Course Assistant* *Stanford, CA*

- Courses Taught: Math 18 (**Foundations for Calculus**), Math 113 (**Linear Algebra**), MS&E 232 (**Game Theory**), MS&E 232H (**Accelerated Game Theory**), MS&E 260 (**Introduction to Operations Management**), OIT 272 (**Online Marketplaces, MBA Elective**).

**Columbia University** *2020 – 2022*
*Teaching Assistant* *New York, NY*

- Courses Taught: COMS 3261 (Computer Science Theory), MATH GU4061 (Modern Analysis I), MATH GU4032 (Fourier Analysis), MATH UN3028 (Partial Differential Equations), MATH UN2010 (Linear Algebra), MATH UN1205 (Accelerated Multivariable Calculus).

**One-to-One Tutoring, Columbia University** *2020 – 2022*
*Tutor* *New York, NY*

- Math tutor for elementary school students, focusing on improving skills and study habits.

## Awards and Recognition

**Lillie Fund** (2026), $16K

**Stanford HAI Data Science Scholarship** (2024-2026), $128K

**NSF Graduate Research Fellowship** (2022-2027), $159K

**Stanford EDGE Doctoral Fellowship** (March 2022)

**Columbia Provost's Diversity Fellowship** (March 2022)

**UC Berkeley Chancellor's Fellowship** (February 2022)

## Talks

**"One-sided Randomization in Two-sided Platform Experiments: Which Side Should be Randomized?"**

CODE at MIT 2026 November 2026

**"Can Labor Markets Function in the Age of AI? The Evaluation Bottleneck in Hiring"**

INFORMS 2026, San Francisco October 2026
Stanford HAI Fall Conference 2026 October 2026

**"When Does Interference Matter? Decision-Making in Platform Experiments."**

Economics and Computation 2026, Rome July 2026
CODE at MIT 2025 November 2025
INFORMS 2025, Atlanta October 2025
Experimentation in Operations Workshop, Harvard Business School May 2025
Data-Driven Decisions Seminar, Stanford GSB May 2025

**"False Positives in A/A Experiments with Correlated Outcomes."**

INFORMS 2024, Seattle — October 2024
CODE at MIT 2024 — October 2024
Stanford Causal Science Center Conference — October 2024

**"Dynamic Contracting for Payment for Ecosystem Services Programs."**

MSOM 2024, Minneapolis — July 2024

## Skills

**Languages:** Python, Julia, R, MATLAB, Java, C++.

**Libraries/Technologies:** NumPy, Matplotlib, Scikit-learn, cvxpy, Pandas.

## References

**Itai Ashlagi** *Stanford University*
*Professor of Management Science and Engineering*

iashlagi@stanford.edu

**Ramesh Johari** *Stanford University*
*Professor of Management Science and Engineering*

rjohari@stanford.edu

**Irene Lo** *Stanford University*
*Assistant Professor of Management Science and Engineering*

ilo@stanford.edu

**Gabriel Weintraub** *University of Chile/Columbia Business School*
*Professor of Economics and Business*

gweintra1@gmail.com